Pokémon: Shiny Vice — Concept Document

Working draft · October 20, 2026 · Author: Vice

> **Placeholder notice:** every rate, cap, price and number in this document is a placeholder. If the math stops working once all the odds boosts are stacked together, the numbers change, starting with the base rate. The structure matters more than the values.

Status tags used below: **Decided** (stated by Vice), **Proposed** (suggested, not yet confirmed), **Open** (needs a decision). Sections 2 and 4 (pillars and core loop) were synthesized from Vice's notes and the design conversation, so they are Proposed until confirmed.

1. Vision

A Pokémon ROM hack built from the ground up for shiny hunters ("shunters"). It takes Radical Red as its baseline (difficulty, options, quality of life, DexNav-style hunting), blends in mechanics from PokeMMO (hordes, seasons, a visible encounter counter), and adds custom shunting systems so that shiny hunting is the key mechanic and the highlight of the game, not a side activity.

2. Design pillars

1. **Shiny hunting is the core loop.** Every major system either helps you hunt, rewards you for hunting, or makes the result of a hunt worth having.
2. **Transparent odds.** Players can always see their encounter totals, current odds, and how those odds were built up.
3. **Rare, but earned.** Many things raise your odds, but a hard cap keeps shinies rare. Balance is the project's central design problem.
4. **Quality of life everywhere.** QoL applies to every part of the game, not just hunting.
5. **Player choice in grind.** Players who love breeding and EV training get full depth; players who don't can opt into a minimal-grind mode.
6. **Radical Red-style difficulty.** Hard content stays hard, and it is a source of the end-game currency.

3. Foundation and scope

- **Base (Decided, “most likely”):** the pokeemerald decomp together with pokeemerald-expansion.
- **V1 scope (Decided, “probably”):** Generation 1–3 Pokémon only.
- **Radical Red as design reference (Proposed approach):** take the design cues, not the code or assets. Features are reimplemented on the decomp with original data and art. “Radical Red baseline” should mean feature parity, not content parity.
- **Hunting speed (Decided):** no built-in animation skipping or instant flee. Players on emulators are expected to use fast-forward, as they already do in Radical Red.
- **Platform:** a GBA ROM hack that runs on emulators (for example mGBA) and compatible hardware.

4. The core loop

1. Hunt shinies using hordes, chain hunting, and charms.
2. Breed and EV-train them into strong Pokémon.
3. Take them into hard content.
4. Earn Shiny Shards from that hard content.
5. Spend shards on charms, PP Max, pity upgrades and other shunting items, then hunt better.

5. Shiny odds system

5.1 Base rate (placeholder)

Proposed default: **1/4096** per encounter (Decided as a starting point, “possibly higher if needed”). Each encounter is an independent roll, like a coin with 4096 sides: the odds never “build up” on their own, and the expected number of attempts is only an average, with a wide spread around it.

5.2 Roll-budget model (Proposed)

Every boost adds extra shiny rolls rather than a percentage. Total rolls are clamped at a hard cap, which gives one clean balancing knob. With n total rolls, odds are about n/4096 at these rates. Every shiny roll in the game goes through one central function, so charms, chains, breeding and the cap all feed a single calculation.

5.3 Illustrative progression

Balance in expected attempts per shiny, not in minutes, because fast-forward changes real time per attempt.

Stage	Total rolls	Odds	Expected attempts
Early game	1	1/4096	~4,096
Charm tiers 1–3	2–4	1/2048 – 1/1024	~2,048 – ~1,024
Chain bonus (up to +3)	up to 7	~1/585	~585
Cap	8	1/512	~512

5.4 Rules of the model

- **Hordes never chain (Decided).** Hordes are a separate way to shunt. (Proposed, not yet decided: hordes use base odds plus charm only, with a conservative cap, because a horde multiplies the number of rolls per battle.)
- **Chain hunting is for single encounters only (Decided).** Doubles might come later, but Vice doubts they are a good idea.
- **Hard cap on stacking (Decided).** Chain, charms and other sources can never push total odds above the cap.
- **Display (Proposed):** the encounter counter shows the cumulative probability of having found a shiny by now.
- **Bad-luck tail:** because the rolls are independent, a meaningful share of players will go well past the average. Shard-based bad-luck protection is worth considering for wild hunts.

6. Hordes (x3 and x5)

Inspired by PokeMMO hordes. A horde is a group of three or five wild Pokémon fought at once; in PokeMMO, Poké Balls are locked until only one remains, and Sweet Scent always attracts a horde in locations that support them (some locations never spawn hordes). Open: whether the hack adopts the ball lock.

- **Spawning (Decided):** a low chance in certain wild encounters, otherwise triggered by Sweet Scent. Not every route has hordes without Sweet Scent. (Open: the natural spawn chance for each encounter type.)

- **Uses (Decided):** shunting (up to 5 encounters in one battle), plus EV and XP training with multi-target moves.
- **Rotation (Decided):** hordes rotate by location or region, time of day, and season. Because of the rotation, some Pokémon are single-encounter only.
- **Sweet Scent PP (Decided):** Sweet Scent costs PP, as in PokeMMO. PP Max is purchasable with Shiny Shards. (Open: whether PP restores are cheap or shard-gated, since PP is effectively the throttle on horde hunting.)
- **Technical risk:** the Gen 3 battle engine assumes four battlers and derives player/opponent side from battler position. A 1v3 or 1v5 battle needs real surgery on battle logic, UI and targeting. Check for existing community horde implementations before building. (Open: ship x3 first, then x5.)

7. Hunt Navigator (DexNav-style tool)

Working name: **TBD.** Ideas so far: “Huntdex”, “Scent Log”. It is modeled on the Radical Red DexNav but customized.

An encounter viewer first (Decided). It shows every encounter in your current area for the current day, night and season, and tags the encounter types: single, double, horde and so on. Unknown species appear as silhouettes (Decided).

Chain hunting is one function of it (Decided). If you have caught a species before, you can select it and begin chain hunting it. Locking in a target costs Shiny Shards. Chains go up to about x100 and raise your odds incrementally within the global cap.

Proposed viewer features: real slot rates, caught and shiny markers, rotation countdowns (“Fall begins in 9 days”), a method filter (grass, surf, fishing, cave), a marker for single-encounter-only species, and horde tags with Sweet Scent versus natural spawns.

Open decisions:

- What breaks a chain: fleeing, KOing the target, catching something else, leaving the area, switching targets?
- Does the chain persist across saving and quitting?
- Lock-in cost structure: a flat cost, a partial refund on catch, or cost tied to chain length.
- Bonus curve: placeholder of about +1 roll per 25 chain, up to +3 or +4 at x100. Vice’s notes say odds rise “depending on your current rate”, so also decide between additive rolls (the current model) and a multiplier on current odds.
- Extra high-chain perks (IVs, abilities, egg moves) versus protecting the breeding loop.

8. Shiny Shards and the economy

The currency name is TBA (“Shiny Shards” is the working name).

- **Sources (Decided):** farmable, tied to end-game or harder content, with many different ways to obtain them. Proposed candidates: Battle Frontier-style challenges (an Emerald base has the Battle Frontier), hard-mode rematches, legendary encounters, shiny dex milestones, and breeding milestones. Open: the final list.
- **Sinks (Decided):** chain lock-ins; tiered shiny charms (an end-game grind, each tier slightly better than the last; Open: number of tiers and their costs); PP Max; other end-game items that aid shunting and breeding; and possibly berry-farming items.
- **Design care:** shards must not require grinding non-shiny content, and every odds source must stay inside the global cap.

9. Breeding and pity

Shiny breeding is a major part of the hack (Decided). The idea is a custom pity system with a GUI players can see, possibly tied into the shard economy (Open: the exact mechanism).

- **Open:** soft ramp (each non-shiny egg adds rolls toward the cap, reset on a shiny) versus hard guarantee at a set number of eggs. A guarantee is easy to display but makes shinies a known quantity.
- **Rule of thumb (Proposed):** shiny breeding should not depend on IV breeding, so it still works in minimal-grind mode.
- Wild hunts could skip pity entirely.

10. Pokémon quality and grind modes

- **Shinies are good Pokémon (Decided):** they spawn with strong IVs. Proposed: a guaranteed number of 31 IVs applied at the moment a shiny is created. (Open: for all sources, or only wild.)
- **IVs and EVs matter (Decided)** for players who enjoy breeding and EV training.
- **Minimal-grind option (Decided):** FireRed-style, with every Pokémon at perfect 31 IVs and no EVs, or a new custom balance. Proposed: a mode chosen at new game (Classic, Casual, Custom).

11. Follower Pokémon

Follower sprites are a must (Decided), and unlike Radical Red they must include shiny variants, so a shiny Pokémon visibly follows you.

Technical note: a shiny follower can be one extra 16-color palette per species instead of a second sprite set. The real cost is sourcing overworld sprites for the roster.

12. Day/night and seasons

- **Day and night cycle with seasons (Decided).** Seasons rotate on the first of each month in PokeMMO (spring, summer, fall, winter). Open: confirm the hack uses the same schedule.
- **Proposed implementation:** drive time and date from the RTC. For V1, do seasons as palette shifts plus swapped encounter tables, and revisit tile art later.
- **Open:** a way to choose the season (cheat terminal or shard purchase) so hunters are not stuck waiting a month for a species to rotate in.

13. Quality of life

QoL is a key point of the project (Decided) and applies to all facets of the game.

- **Encounter counter (Decided):** PokeMMO-style, a visual representation of your odds and total encounters. Proposed: persists across soft resets, per species and per method.
- **No built-in animation skipping or instant flee (Decided).**
- **Open:** the full QoL list. The Radical Red set is the baseline to review: keep, cut, or add.

14. Cheat terminal

A PC in the player's bedroom, like Radical Red, with options such as DexAll, so players who don't care about the Pokédex can make shunting easier from the start (Decided).

- **Proposed:** keep a persistent per-category "cheats used" flag, and mark shiny log and hunt tracker entries as assisted.
- **Candidates:** DexAll, set time of day, set season, rare candies, money, and the minimal-grind options.

- **Open:** shard cheats undercut the end-game grind, so they should be flagged or left out.
- **Recommendation (Proposed):** no cheat changes shiny odds directly. The roll cap is the balance contract. A “give me this Pokémon” cheat, flagged as assisted, covers players who just want the result.

15. Difficulty

Radical Red’s difficulty is a core attraction (Decided). Proposed: let trainers and hard content carry the difficulty while wild encounters stay manageable, so long hunts don’t risk wiping on routine fights. Hard content is also the main source of shards.

16. Story

Undecided. It has to be good. Options: stay close to the original games with added flair, or go fully custom. An Emerald base means a Hoenn-style world and story rather than Radical Red’s Kanto.

17. Technical notes

- **Central shiny roll function:** all odds sources feed one function.
- **Single encounter data source:** extend the decomp’s encounter JSON with season and horde fields, and generate both runtime tables and the Hunt Navigator data from it.
- **Save space:** Gen 3 saves use fixed-size sectors. Counters for each species, chain state, pity meters, season data and cheat flags all need budgeting early.
- **RNG:** Gen 3 RNG is a seeded algorithm. It behaves independently in normal play, but soft-reset hunts can interact with the seed. Decide early whether to harden it, expose it, or make it a toggle.
- **Tech to verify:** current status of DexNav ports, follower support, and time-of-day encounters in pokeemerald-expansion.

18. Open decisions

1. What breaks a chain, and does it persist through saves?
2. Chain lock-in cost, the bonus curve, and whether chain bonuses are additive rolls or a multiplier on current odds.

3. Horde cap level, whether to ship x3 before x5, the natural horde spawn chance, and whether the Poké Ball lock carries over.
4. Sweet Scent PP restore cost: soft or hard throttle.
5. Breeding pity: soft ramp or hard guarantee, and how it connects to shards.
6. Do shiny IV guarantees apply to eggs, or only wild catches?
7. The final list of shard sources, and whether shard cheats are allowed and flagged or excluded.
8. Number of shiny charm tiers and their costs.
9. Non-wild hunts (starters, gifts, static legendaries, fishing): which get chains and charm bonuses, and which are reset hunts.
10. Season schedule and the season selection method for hunters.
11. The full QoL list.
12. Story direction.
13. Names: the currency and the Hunt Navigator.
14. Distribution approach: patch-based release and original assets (not legal advice; worth settling before going public).

19. Suggested build order

1. Repo setup, base shiny odds, and the encounter counter, to validate the pipeline.
2. Roll budget and cap as a single pure function.
3. Hunt Navigator with chain hunting and the shard currency.
4. Hordes.
5. Breeding pity.
6. Followers (including shiny) and seasons.

20. References and inspirations

- PokeMMO Horde page: pokemmo.shoutwiki.com/wiki/Horde
- Inspirations: PokeMMO (about 200 hours played), Radical Red, and Unbound (admired for breaking the limits of earlier hacks).
- Calibration data point: a shiny starter Litten in Radical Red in under 4 hours, soft-resetting on mGBA with fast-forward.